

Let's take an example of the impact of a single social media post. A real-time report that tracks social media mentions; a sentiment analysis tool that gives a good, negative, or neutral conclusion; or an app charting each mention on a globe map might all be fed with data from this event. Despite sharing a common data source, these apps all rely on different data pipelines that must run without a hitch to produce the desired output for the end user.

Data transformation, enhancement, filtering, grouping, aggregation, and the application of algorithms to data are all typical components of data pipelines.

Pipelines in Big Data scenarios

The term Big Data indicates that there is a large amount of data to process, and this is true because the volume, variety, and velocity of data have all increased considerably in recent years. Having access to all this information can be useful for a wide variety of purposes.

Data pipelines have progressed with other parts of data architecture to accommodate huge data. A compelling reason to construct streaming data

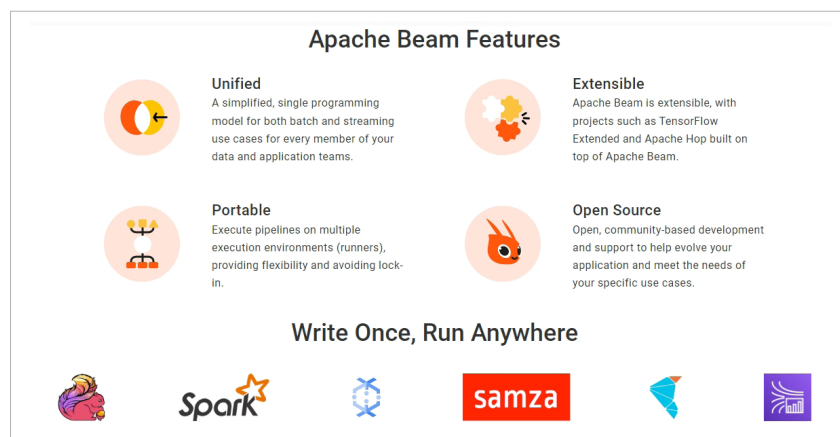


Figure 3: Key features and integration libraries with Apache Beam

pipelines for Big Data is the volume and speed of this data. These pipelines help to gather information in real-time and act upon it.

The Big Data pipeline needs to be scalable in order to process massive amounts of data concurrently, as there are likely to be numerous data events that occur at the same time or very near together. The pipeline must also be able to detect and process heterogeneous data, including structured, unstructured, and semi-structured data.

Working with Apache Beam

Apache Beam (<https://beam.apache.org/>) is a free and open source unified programming model for describing and working with effective data processing pipelines. This data processing methodology is used to work with the pipelining and real-time filtering of data for multiple applications. All data sets and data frames can be processed using a single API with the integration of Apache Beam. To 'build and run in any place' is a key concept of Apache Beam.

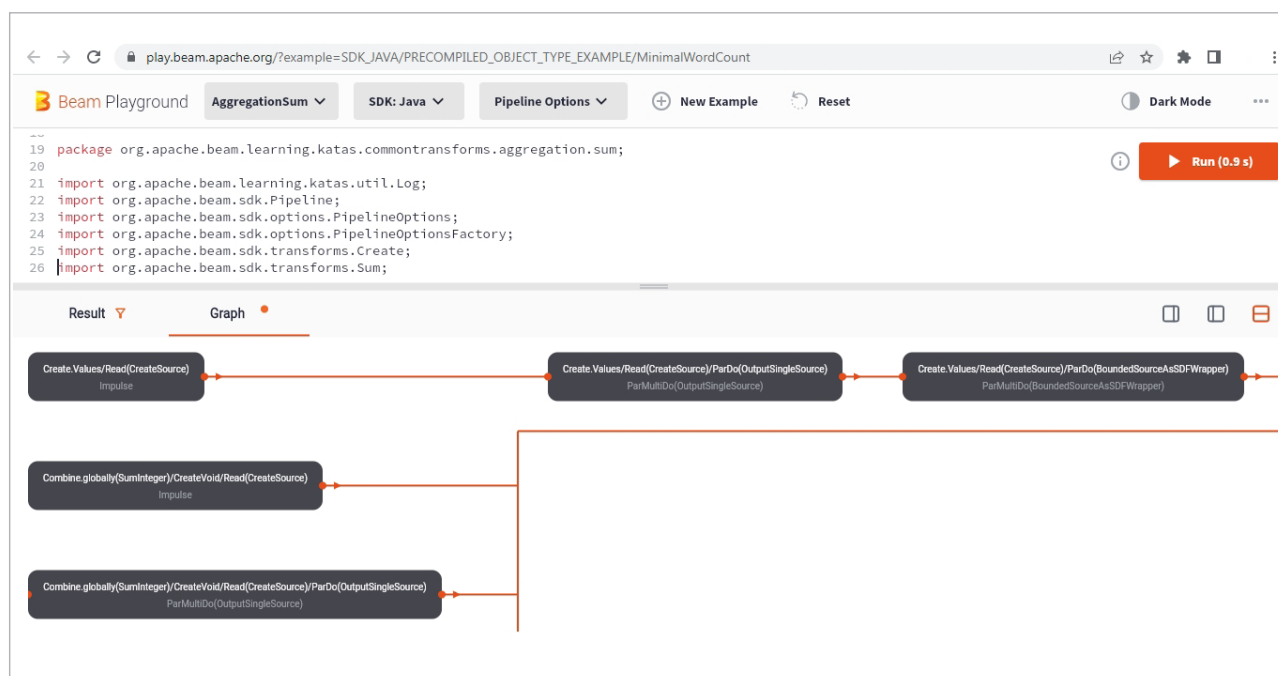


Figure 4: Apache Beam Playground